Lucas Neumann

Senior Software Engineer — Pricing, Risk & AI Tooling

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Summary

Five years building pricing and risk systems on both sides of the market: an investment manager, Bank of America's FICC emerging markets desk and a hedge fund. Rates, credit and FX derivatives, Bloomberg MARS, vendor PMS platforms, VaR, expected shortfall and Greeks — backed by Python, Go and cloud engineering.

Experience

BWGI · Senior Software Engineer — 2025–present

- Took **Bloomberg MARS** front-office pricing from proof of concept to daily production, delivering better-quality prices for product classes the legacy PMS did not cover and removing lock-in to that vendor. Automated end to end (portfolio generation, SFTP/S3 delivery, report ingestion) with regression tests and Datadog monitoring.
- Built MARS mappings for rates, credit, FX and fixed-income instruments — IRS (cleared, OIS, CPI), CDS, TRS, FX forwards/NDFs, barrier and digital options, bonds, CDs and repos — handling quote conventions, face-amount scaling and composite tickers.
- Integrated Bloomberg (MARS, Data License), Arcesium PMS and other data vendors through ETL pipelines covering 20+ funds and thousands of instruments daily (3–5 GB per portfolio), reconciled automatically against legacy pricing.
- Designed a new risk data pipeline — risk-engine output → PostgreSQL → Qlik — with Greeks and risk-factor serving tables and a data-quality gate (cross-layer reconciliation, robust z-score outliers), run daily per fund.
- Primary on-call for production pricing and risk systems.

Bank of America · Software Engineer, AVP — 2023–2025

FICC Emerging Markets. Joined via BRQ as consultant in 2023; hired directly as AVP in 2024.

- Built and ran real-time curve infrastructure for the EM derivatives and FX trading desk: live market data and trader curve marks reaching the sell-side pricing system **within seconds**, plus intraday and EOD curve publication to internal databases; onboarded new engineers to the curve stack.
- Took the real-time **PTAX** integration with Banco Central do Brasil to production and kept it stable — a high-availability service where any failure carries direct reputational risk.
- Maintained rates and inflation curve frameworks and bond pricing tools for FX and derivatives traders, resolving production issues with the desk day to day.

Apex Capital · Software Developer — 2021–2022

Hedge fund

- Cross-area role, working day to day with research, economics and operations: database administration, data analysis and research reports, web scraping for data mining, Streamlit risk dashboards and automation of Excel-based operations workflows.
- Migrated the fund's scheduled jobs from on-premises Task Scheduler to Airflow on AWS, with version control, CI, tests and Docker.

Skills

Engineering
Python, Go, Rust, SQL (PostgreSQL, Snowflake), Airflow/Astronomer, MongoDB, Redis, Docker, Kubernetes, AWS, Datadog, MCP

Financial systems
Bloomberg MARS & Data License, Arcesium, MSCI Barra, S&P Global, Qlik

Domain
Rates, credit, FX, fixed income; derivatives pricing; VaR, expected shortfall, Greeks

Languages
Portuguese (native), English (fluent)

Education

UNICAMP — University of Campinas
MSc, Physical Chemistry
2018–2020
Computational simulation of proteins; daily work with HPC clusters, Python, Julia, C, C++ and Linux.

Projects

git-archaeologist (Rust)
Fast CLI extracting time series from git history (LOC, cohort survival, coupling, hotspots) as TSV, CSV, JSON or Parquet.

visidata-rs (Rust)
Rewrite of the VisiData terminal data explorer in Rust (Polars, Ratatui), Lua-extensible like Neovim; in progress.

StringArt.jl (Julia)
Generative art: turns photos into string-art patterns.

MonteCarloArt.jl (Julia)
Generative art: pointillist images via Monte Carlo sampling.